EE257 Pre-Lab

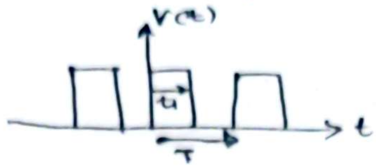
Basic Operation of a Spectrum Analyzer and Time-frequency Domain conversion of signals.

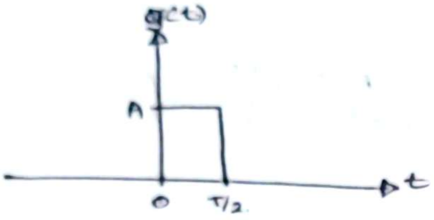
E/21/345 Samarakoon S.M.O.T.

Checked

① Calculate the fourier coefficients of following periodic signal & then determine its power spectrum in terms of A

for → a) $T = 2\mu s$
b) $T = 4\mu s$.





assume, $t_1 = \frac{T}{2}$.

$$g(t) = \begin{cases} A & ; 0 \leq t < T/2 \\ 0 & ; \text{elsewhere} \end{cases}$$

$$v(t) = \sum_{k=-\infty}^{\infty} g(t - kT) \quad ; k \in \mathbb{Z}$$

$T_0 = T$

$w_0 = \frac{2\pi}{T}$

Fourier coefficients of this X_k.

$k = 0$,

$$X_0 = \frac{1}{T_0} \int_{T_0} v(t) (e^{-j(0)w_0 t}) dt$$

$$= \frac{1}{T} \int_0^{T/2} A(1)\, dt$$

$$= \frac{1}{T} (At) \Big|_0^{T/2} = \underline{\underline{\frac{A}{2}}}$$

when $k \neq 0$,

$$X_k = \frac{1}{T_0} \int_{T_0} v(t)\, e^{-jkw_0 t} dt$$

$$= \frac{1}{T} \int_0^{T/2} A\, e^{-jkw_0 t} dt$$

$$= \frac{A}{T} \left(\frac{e^{-jkw_0 t}}{-jkw_0} \Big|_0^{T/2} \right)$$

$$= \frac{A}{-jkw_0 T} \left[e^{-jk\pi} - 1 \right]$$

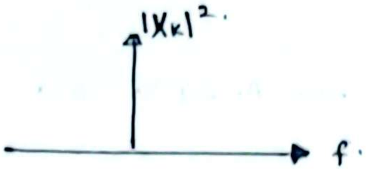
$$= \underline{\underline{\frac{Aj}{2\pi k} \left[e^{-jk\pi} - 1 \right]}}$$

$$\text{fourier coefficients} = X_k = \begin{cases} A/2 & ; k = 0 \\ \frac{Aj}{2\pi k} (e^{-jk\pi} - 1) & ; k \neq 0 \end{cases}$$

$$\underline{v(t) = \frac{A}{2} + \sum_{\substack{k=-\infty \\ (k \neq 0)}}^{\infty} \left[\frac{Aj (e^{-jk\pi} - 1)}{2\pi k} \right] e^{jkw_0 t}}$$

↑ fourier series.

(a.)



$T = 2\mu s$

$\omega_k = 2\pi f_k$

$$f_k = k f_0 = \frac{k}{T_0} = \frac{k}{2\times10^{-6}} = \frac{k}{2}\ MHz$$

$$X_k = \frac{Aj}{2\pi k}\left(e^{-jk\pi} - 1\right)$$

$k=1$; $X_1 = \frac{Aj}{2\pi}\left(e^{-j\pi}-1\right) \longrightarrow |X_1|^2 = \left|\frac{A}{2\pi}(-2j)\right|^2 = \frac{A^2}{\pi^2}$

$k=2$; $X_2 = \frac{Aj}{4\pi}\left(e^{-j2\pi}-1\right) \longrightarrow |X_2|^2 = \left|\frac{A}{2\pi}(0)\right|^2 = 0$

$k=3$; $X_3 = \frac{Aj}{6\pi}\left(e^{-j3\pi}-1\right) \longrightarrow |X_3|^2 = \left|\frac{A}{6\pi}(-2j)\right|^2 = \frac{A^2}{9\pi^2}$

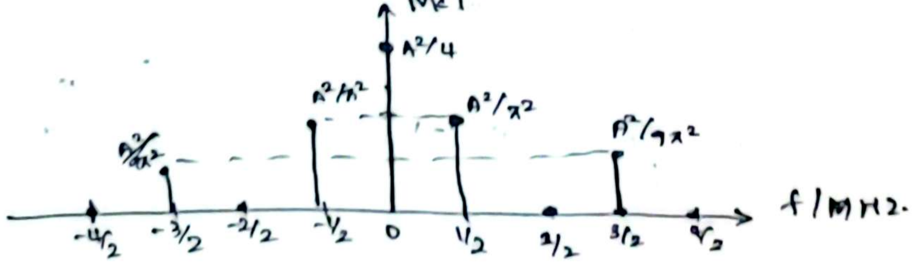
$k=-1$; $X_{-1} = \frac{Aj}{-2\pi}\left(e^{j\pi}-1\right) \longrightarrow |X_{-1}|^2 = \left|\frac{A}{-2\pi}(-2j)\right|^2 = \frac{A^2}{\pi^2}$

$k=-2$; $X_{-2} = \frac{Aj}{-4\pi}\left(e^{j2\pi}-1\right) \longrightarrow |X_{-2}|^2 = \left|\frac{A}{-4\pi}(0)\right|^2 = 0$

$k=-3$; $X_{-3} = \frac{Aj}{-6\pi}\left(e^{j3\pi}-1\right) \longrightarrow |X_{-3}|^2 = \left|\frac{A}{-6\pi}(-2j)\right|^2 = \frac{A^2}{9\pi^2}$

power spectrum $|X_k|^2 = \begin{cases} \frac{A^2}{(|k|\pi)^2} & ;\text{ if } k \text{ is odd} \\ 0 & ;\text{ if } k \text{ is even} \end{cases}$

for $k=0 \rightarrow |X_0|^2 = \frac{A^2}{4}$



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(b.) $T = 10\mu s$,

$$f_k = \frac{k}{T} = \frac{k}{10 \times 10^{-6}} = \frac{k}{10} \text{ MHz}$$

likewise,

power spectrum $\rightarrow$ $|X_k|^2 = \begin{cases} \frac{A^2}{(|k|\pi)^2} & ; \text{ if } k \text{ is odd} \\ 0 & ; \text{ if } k \text{ is even.} \end{cases}$